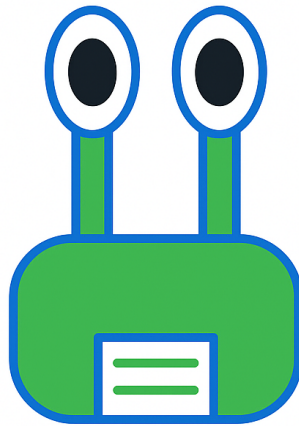


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2025**



ByteBuddi

Build Better Resumes

**TEAM BYTEBUDDI
AUTOMATED RESUME BUILDER**

**RYAN DAVIS
RENE SALAZAR
ADALYNN EVERLY
CESAR GALLEGOS
ALAN ACEVEDO
XAVIOR LOPEZ**

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	9.15.2025	RD, RS	Document Creation
0.2	9.26.2025	RD, AE, RS, AA, XL, CG	Complete Draft
0.3			
1.0			
1.1			

CONTENTS

1 Problem Statement	6
2 Methodology	6
3 Value Proposition	6
4 Development Milestones	6
5 Background	7
6 Related Work	7
7 System Overview	8
8 Roles & Responsibilities	8
9 Cost Proposal	9
9.1 Preliminary Budget	9
9.2 Current & Pending Support	10
10 Facilities & Equipment	10
11 Assumptions	10
12 Constraints	10
13 Risks	11
14 Documentation & Reporting	11
14.1 Major Documentation Deliverables	11
14.1.1 Project Charter	11
14.1.2 System Requirements Specification	11
14.1.3 Architectural Design Specification	11
14.1.4 Detailed Design Specification	11
14.2 Recurring Sprint Items	11
14.2.1 Product Backlog	11
14.2.2 Sprint Planning	11
14.2.3 Sprint Goal	12
14.2.4 Sprint Backlog	12
14.2.5 Task Breakdown	12
14.2.6 Sprint Burn Down Charts	12
14.2.7 Sprint Retrospective	12
14.2.8 Individual Status Reports	12
14.2.9 Engineering Notebooks	13
14.3 Closeout Materials	13
14.3.1 System Prototype	13
14.3.2 Project Poster	13
14.3.3 Web Page	13

14.3.4 Demo Video	13
14.3.5 Source Code	13
14.3.6 Source Code Documentation	13
14.3.7 Installation Scripts	13
14.3.8 User Manual	14

LIST OF FIGURES

1	Example Function Overview	8
2	Example sprint burn down chart	12

1 PROBLEM STATEMENT

The average student has minimal experience in crafting a resume and lacks a tool or process to help guide them in creating an impactful resume for potential employers or internships. Our goal is to provide UTA students a resource to help them create professional and effective resumes that they can be confident in as they pursue potential internships or positions outside of their college careers.

2 METHODOLOGY

We are dedicated to developing a resource students can use in their web browsers to create a perfect resume to fit their internship goals. We plan on having a user experience that helps students seamlessly create a stable resume from a variety of templates offered in our curated selection. The resume creation will also be packaged with a feature for AI generated interview questions, further preparing students for their internship roles.

Some implementation methods we want to provide students are as follows:

- Account creation and database storage
- Resume PDF uploading and downloading
- Editing of resumes
- Generation of resumes from templates
- AI mock interviews

3 VALUE PROPOSITION

We believe that there is value in preparing students for the corporate world. Giving them the tools necessary to create resumes and prepare them for interviews gives students an invaluable level of confidence that they can carry on to their future internships and corporate jobs.

Companies can benefit from hiring students who have clear visuals of their experiences and achievements in professional formats within their resumes. These candidates will also be prepared

4 DEVELOPMENT MILESTONES

- Project Charter first draft - September 26th, 2025
- System Requirements Specification - October 17th, 2025
- Architectural Design Specification - November 7th, 2025
- Project Gate Review - December 1st, 2025
- Detailed Design Specification - TBD 2026
- CoE Innovation Day poster presentation - TBD 2026
- Final Project Demonstration - May 2026

5 BACKGROUND

Students at UTA are offered "Resume Workshops" at various times throughout the year; however, these workshops are only available for limited times and not every student can fit that time into their busy school schedules. We aim to provide UTA students with a resource that they can use a resource that they can use year-round to perfect their resumes and prepare for any upcoming interview questions they may face. Providing students with professional, curated templates that are formatted for their specific work field can help them feel confident in themselves and the skills they can bring to a hiring company or internship. Once students have selected and created their own professional resume, they can further prepare themselves by engaging with our on-site AI mock interview feature. This feature is dedicated to help students get practice with real-world scenarios and questions that they may face in their future interviews. Empowering students here at UTA with tool to help them land their dream internships or jobs is what our project strives to achieve.

6 RELATED WORK

The current state-of-the-art with respect to our product includes services such as Zety [1], Resume.io [2], and ResumeNow. [3]. The technologies used to develop similar products to ours consist of frontend frameworks using JavaScript [4] and React [5] while backend uses Ruby [6] or Python [7] with MySQL [8] handling database queries. While all of these languages and frameworks are open source and widely available, our competitors offer their services at premium prices.

Zety.com is a resume builder website known as a professional career site that offers a suite of powerful tools that are designed to help make the job search process simpler and more effective for users. They have several templates to choose from, also offer cover letter templates and cover letter builders. Zety offers a free subscription package that has access to almost everything, the only downside of this package is that you don't get access to downloading your resume, that option is available on the other subscription packages one that starts at \$25.95 monthly or the annual one that starts at \$71.40 for a one time pay or a monthly pay of \$5.95. [1].

Resume.io is an AI-powered resume builder that helps you create a resume, or if you already have one, it will tailor it to the job you are trying to apply to. It also offers cover letters that are generated by AI, it supports several speaking languages around the world that way it's something anyone around the world has access to. The only downside about this company is that it's not free. Resume.io offers two subscription plans, the free trial that starts at \$2.95 then after the free trial is over you are on a monthly charge of \$29.95. They also have a quarterly charge that starts at \$49.95 which is every three months, both plans offer the same benefits. The cost of the plans for an entire year ranges from \$200 - \$340 which is really expensive for students. [2].

Lastly, we will talk about ResumeNow which is another AI resume builder this one is considered to be fast, easy, and free to use. It will help create a resume from scratch by choosing one of the approved templates, from there you will add your skills and strongest points. It will then generate a resume for you in a matter of minutes that will be available in a PDF format. It also offers built in cover letters and a section for advice from other Resumes, Cover Letters, Job Search, or Career Advice. After creating an account, looks like to receive all the benefits we described you will need to purchase a subscription that starts at \$23.85 monthly or a yearly one-time payment of \$70.20. The only thing that is free from this website is that it allows you to have access to a certain amount of templates and a limited attempt of corrections and guidance. [3].

One thing to keep in mind is that each company offered a subscription plan that was at least \$25 monthly or fewer than \$100 for a yearly subscription. What we are trying to focus on is by making sure that this design is free for students at the university, that way they can avoid these sites and will not have to pay for a subscription, we are also going to be offering mock interviews that are built in with

AI, which none of these companies had to offer. They only offered AI to adjust their resume, but not prepare them for interviews.

7 SYSTEM OVERVIEW

Our system will rely on both the interaction of our users on the web page and continuous calls to a database. On the frontend, we will be working with both React and Next.js. Backend will be structured with FastAPI(Python) coupled with integration into the FireBase database system. Lastly, we will be utilizing OpenAI/HuggingFace APIs for our mock interview features.

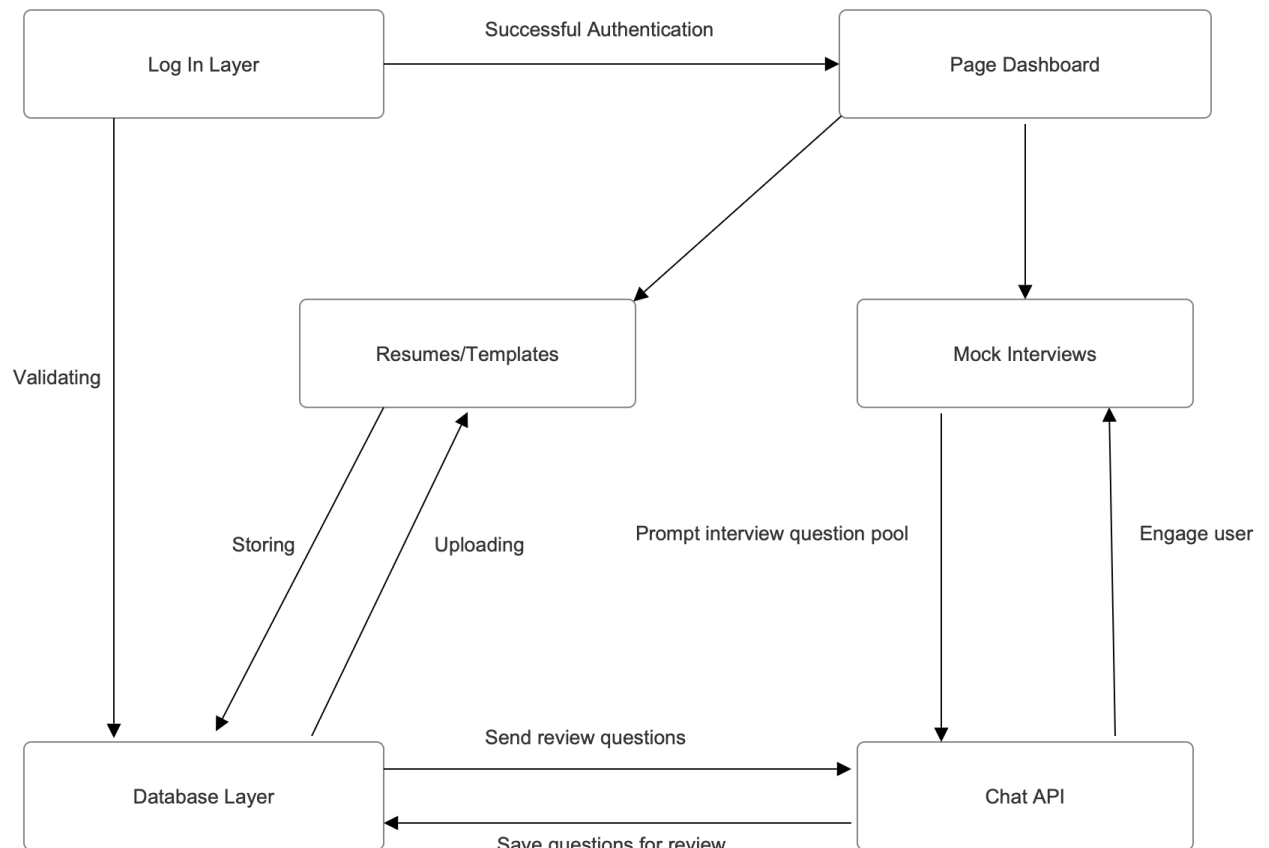


Figure 1: Example Function Overview

8 ROLES & RESPONSIBILITIES

The stakeholders for our product include UTA, the instructor, and all the individuals working on the product. UTA is concerned with the quality of the product by making sure that all requirements are met. The student base of UTA will be positively impacted by the successful deployment of ByteBuddi. The instructor is concerned with the quality of work in the design, development, and documentation

of the product. The team behind the ByteBuddi project is affected on the basis of the quality of their work. The team is concerned with making prompt deliveries on sprint deadlines, taking care during documentation, and developing a successful product while adhering to the product requirements.

Throughout the project's lifetime, our team will work cohesively on bigger project deliverables. However, we have structured roles in our team to expedite workflow among team strengths and positions:

- Adalynn (Product Owner)
 - Owns the backlog, requirements, and priorities. Acts as main link between team and instructor.
- Ryan (Scrum Master)
 - Runs standups, retros, and burn-downs. Also responsible for Overleaf management and keeping docs consistent.
- Alan (Firebase Setup and Cloud Config)
 - Sets up and maintains Firebase (Firestore + Auth + Storage). Handles hosting integrations and environment variables.
- Rene (Backend Development)
 - Works on core backend logic, APIs, and integration with AI models (OpenAI/HuggingFace).
- Cesar (Frontend Development)
 - Focuses on frontend UI, resume builder interface, and interview page.
- Xavier (Resume Templates and Export Features)
 - Designs and implements resume templates, PDF generation, and export/download functionality.

9 COST PROPOSAL

The starting budget granted by UTA is the funding that our team will be working with, which is more than enough for the development of our product. This starting budget of \$800 will be used to pay for a website domain, database features and storage, plus any other potential software expenses.

9.1 PRELIMINARY BUDGET

List of potential expenses related to project requirements

Items	Amount \$
Website Domain	20
Firebase Cost	150
Hosting	60
AI API Credits	200
PDF Storage	20
Misc/Contingency	100
Total	500

9.2 CURRENT & PENDING SUPPORT

Currently the only funding we have access to is the \$800 budget granted by UTA. We do not anticipate needing any further funding from outside sponsors.

	Amount \$
UTA	800
TBD	0

10 FACILITIES & EQUIPMENT

Due to the nature of our product being entirely software, we have minimal equipment requirements to develop our project. Outside of necessary database tools and fees, our team is working with open-source languages and have access to our own computers to develop on remotely. Some of the software and repositories we will be working with include Overleaf for documentation, GitHub for code repositories, and Slack/Discord for team communication.

Should we need a meeting space to accommodate all team members, we have access to meeting rooms that are provided on campus at UTA.

11 ASSUMPTIONS

- The firebase system (Firestore + Auth) will remain stable
- All users will have mavs.uta.edu email
- Project scope will not exceed free-tier/low-tier Firebase limits
- AI API (OpenAI/HuggingFace) will be functional at deployment
- Database security will be stable throughout lifetime of project use
- Team members will have consistent weekly availability

12 CONSTRAINTS

The following key constraints limit the design and implementation of ByteBuddi:

- **Budget:** Total development and deployment costs must not exceed the \$800 provided by UTA. This includes domain registration, Firebase usage, hosting, and API credits.
- **Schedule:** Major deliverables are fixed by the Senior Design course calendar, including the Project Gate Review (December 2025) and final demonstration (May 2026).
- **Technology:** The system must be built using the approved stack of React/Next.js, FastAPI, and Firebase (Auth, Firestore, Storage). Alternative technologies are not permitted without instructor approval.
- **Access Control:** The system must restrict accounts to users with @mavs.uta.edu email addresses for authentication.
- **Data Privacy:** All student data must remain within Firebase services; no personal data can be stored on unapproved external systems.
- **Team Resources:** Development must be completed using only team-owned laptops and freely available software tools; no additional hardware purchases will be made.

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Integrating Firebase Auth across frontend and backend may take longer than estimated.	0.50	4	2
Schema may not support multiple resumes or templates per user, leading to refactors.	0.50	4	2
First-time setup of GitHub Actions and Vercel may fail, preventing live demo.	0.50	3	1.5
Exams and personal commitments reduce availability during key sprint tasks.	0.80	3	2.4
Learning curve for new tools (FastAPI, Next.js, AI APIs)	0.50	4	2
If Firebase rules or schema arenât configured correctly, unauthorized access or leaks could happen	0.50	3	1.5

Table 1: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

14.1.1 PROJECT CHARTER

All 5 team members are collaboratively involved in working on the Project Charter throughout the project timeline, with additions made on individual timelines. The team communicate daily across a Discord server to ensure all parts in the charter are accounted for. The final deliverable is expected to be in Spring 2026.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

All 5 team members will work together to complete the System Requirements Specification document. Our team will check in with each other during the document creation to provide feedback and make progress in completing the document for it's October 17th, 2025 deadline.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

All 5 team members will work together to complete the Architectural Design Specification document. Our team will check in with each other during the document creation to provide feedback and make progress in completing the document for it's November 7th, 2025 deadline.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification is set to be created and delivered in Spring 2026. This document will be created collaboratively in Overleaf among the team.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Items will be added to the Product Backlog in Microsoft Excel as the project progresses along in the SRS. The team will decide which features need to be completed sooner, when others can be pushed back into future sprints. New items/features will be taken into account upon the stakeholders requirements.

14.2.2 SPRINT PLANNING

Sprint planning will be performed at the beginning of each sprint. The team will review what processes have been implemented and which have yet to be done, in order to achieve the project's requirements. There are expected to be between 6 to 8 sprints with each sprint lasting 2 to 3 weeks.

14.2.3 SPRINT GOAL

Our team will set goals for each sprint based on our overall progress within the SRS. Once we have determined our goal for the sprint, we review with our stakeholder to make sure that the requirements are being met throughout the sprint.

14.2.4 SPRINT BACKLOG

The sprint backlog will be maintained in a shared Microsoft Onedrive location for access to Excel and Word. Here, we track what features have yet to be implemented and the required hours worked to reach completion. The team comes together after each sprint cycle to determine which features can make their way to the sprint backlog from the product backlog.

14.2.5 TASK BREAKDOWN

Tasks are broken down in the team with strengths and interests taken into account. If there is no preference, the team's Product Owner and Scrum Master will help guide members into roles that suit them best. The time spent on each role is logged by each team member and reported into a time log excel sheet.

14.2.6 SPRINT BURN DOWN CHARTS

The team will work together to create the burndown charts for each sprint. There is a team excel sheet open to all members for transparent hours worked by each member. Our burndown charts feature a table of hours along with a visual of hours distributed over the sprint.

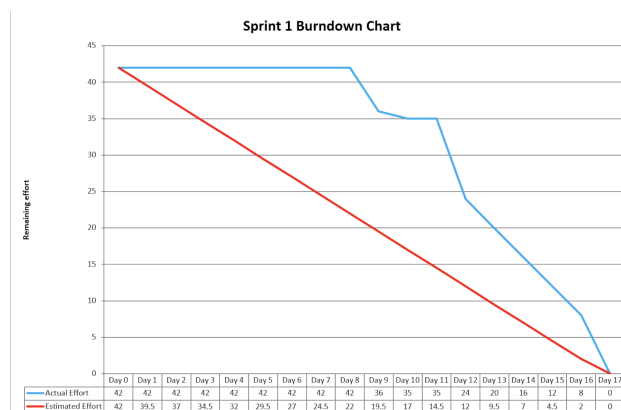


Figure 2: Example sprint burn down chart

14.2.7 SPRINT RETROSPECTIVE

At the end of every 2 weeks, our team comes together and reviews the sprint. We look at what features were implemented and which ones need to be pushed into the next sprint. After we conclude our review, we present our findings to the stakeholder for transparency.

14.2.8 INDIVIDUAL STATUS REPORTS

Each member will complete individual status reports for the sprint that clarify what each member worked on, their upcoming assignments, and how effective each member of the team was during the sprint.

14.2.9 ENGINEERING NOTEBOOKS

Each team member is required to update their engineering notebook at least twice per week, with a minimum of two pages per week. Notebooks will include progress updates, design decisions, and reflections on challenges. Accountability will be checked during weekly meetings. A rotating peer will sign as a witness for each update to confirm accuracy and compliance.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

Before launch, there will be a Prototype Acceptance Test (PAT) with stakeholders featuring our web application in Spring 2026. Features such as account creation, template generation, mock interviews, and PDF uploads/downloads are expected to be presented. Here we will assess reception and any last requirements submitted by our stakeholder before we certify our product.

14.3.2 PROJECT POSTER

The project poster will display visuals of our web application, features that are offered to users, and will address the problem that our application aims to solve. The final dimensions will be subject to UTA student display standards.

14.3.3 WEB PAGE

The entire application will be included on the project web page which is open to all UTA students. We will have a landing page with guides on account creation, resume uploading/downloading, and visual examples of what will be offered to our users.

14.3.4 DEMO VIDEO

The demo video will guide the audience through initial account creation, uploading and editing a resume with our templates, downloading generated resumes, and interacting with the AI mock interview feature. We expect the video to be between 8 - 10 minutes long.

14.3.5 SOURCE CODE

Our source code will be maintained in a team repository on GitHub using Git as the version control. The final source code will be submitted along with a README file containing information on startup, references, and documentation.

14.3.6 SOURCE CODE DOCUMENTATION

We will use JSDoc to document all of our React source code and MDN Webdocs for CSS source code. All of our documentation will be provided as a PDF format for submission to the stakeholder.

14.3.7 INSTALLATION SCRIPTS

ByteBuddi runs in a web browser, so there is no software for the user to install. For anyone setting up the project for development, we will include instructions in the GitHub repository. The repo will have:

- A README file with setup instructions
- npm scripts for frontend dependencies
- Python `requirements.txt` for backend dependencies
- Firebase configuration steps (with placeholders for API keys)

This ensures the system can be rebuilt and deployed by any future developers or evaluators.

14.3.8 USER MANUAL

A short PDF user manual will be provided, including screenshots for login, resume creation, PDF export, and mock interview features. A 2 minute demo video will also be prepared to supplement the manual and walk users through setup and usage. The manual will be written for non-technical users so that the system can be demonstrated and understood without prior training.

REFERENCES

- [1] Zety. "AI Resume Builder: Easy, Fast & Smart Resume". (Accessed: 09.19.2025).
- [2] Resume.io. "Resume Builder: Make & Download Easily". (Accessed: 09.19.2025).
- [3] ResumeNow. "Job-Winning Resume Templates". (Accessed: 09.19.2025).
- [4] JavaScript. "JavaScript". (Accessed: 09.19.2025).
- [5] React. "The library for web and native user interfaces". (Accessed: 09.19.2025).
- [6] Ruby. "Ruby A Programmer's Best Friend". (Accessed: 09.19.2025).
- [7] Python. "Welcome to python.org". (Accessed: 09.19.2025).
- [8] MySQL. "The world's most popular open source database". (Accessed: 09.19.2025).